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Original Contribution

Temperature dependence of ^{1}H NMR chemical shifts and diffusivity of confined ethylammonium nitrate ionic liquid

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ABSTRACT

Some ionic liquids (ILs) change their dynamic properties when placed in a confinement between polar surfaces (Filippov et al., Phys. Chem. Chem. Phys. 2018, 20, 6316). The diffusivities of ions and NMR relaxation times in these ILs also reversibly change under a strong static magnetic field. The mechanisms of these phenomena are not clear, but it has been suggested that they involve modified hydrogen-bonding networks formed in these ILs in the presence of polar surfaces. To obtain a better understanding of these effects, we performed temperature-dependent measurements of chemical shifts and diffusion coefficients for ethylammonium nitrate (EAN) IL in the bulk phase ($\mathbf{I_B}$) and confined in layers with a thickness of ~4 μm between quartz plates unexposed (**I** phase) and exposed ($\mathbf{I_{MF}}$ phase) to a static magnetic field of 9.4 T. It was shown that the NMR chemical shift of NH_3 protons of EAN in the **I** phase is strongly shifted upfield, ~0.0145 ppm/K, which is due to weakening of the hydrogen-bonding network of the confined EAN. Exposure to the magnetic field leads to restitution of the hydrogen-bonding (H-bonding network). The temperature dependences of diffusion coefficients follow the order D(**I**) > D($\mathbf{I_B}$) > D($\mathbf{I_{MF}}$) and can be described by a Vogel-Fulcher-Tammann approach with variation of the pre-exponential factor, which is determined by the strength of the H-bonding network. Confinement of EAN between plates ($\mathbf{I_B} \rightarrow \mathbf{I}$) is an endothermic process, while processes occurring in a magnetic field, $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ and $\mathbf{I_{MF}} \rightarrow \mathbf{I}$, are exothermic and endothermic, respectively.

1. Introduction

Ionic liquids (ILs) are salts prepared from organic cations and either organic or inorganic anions that remain in a liquid state near room temperature. Their design and related properties provide novel potential applications for this new class of material [1–3]. Ethylammonium nitrate (EAN) is the most frequently reported IL [2,4], which is used as a reaction medium, as a precipitating agent for protein separation [5], and as an electrically conductive solvent in electrochemistry [6]. EAN has three readily exchangeable protons on the -NH_3 group of cations; therefore, they belong to a class of so-called "protic" ionic liquids, which are formed by proton transfer from an equimolar combination of a Brønsted acid and a Brønsted base [2]. Like water, EAN has a three-dimensional hydrogen-bonding (H-bonding) network, which is highly connected, dense, complex and also highly variable [7,8]. H-bonding in ILs is a secondary (after ionic bonding) structure-directing contributor [8]. Cations are the dominant H-bond donors, while anions are the dominant H-bond acceptors. The presence of the H-bonding network affects the physicochemical properties of EAN, like many other ILs. The microstructure of protic ILs depends greatly on the hydrogen bond interactions [9]. Changes in intermolecular interactions and electronic perturbations caused by variations in hydrogen bonds can be observed from ^{1}H NMR chemical shifts [9,10]. Typically, higher temperatures result in weakening of the hydrogen bonds and therefore lessen the electron-withdrawing effect of the hydrogen bond acceptor on the proton [10]. As a result, the proton becomes more shielded and its chemical shift value decreases (the resonance moves "upfield"). For example, ^{1}H NMR chemical shifts of alkyl protons of *n*-butylammonium acetate and *n*-butylammonium nitrate are almost unchanged with increasing temperature due to the absence of hydrogen bond interactions between alkyl protons with anions, whereas those of N—H of cations decrease linearly with increasing temperature, indicating that the hydrogen bond interaction weakens gradually [9].

Confined ILs [11–17], particularly EAN [12,15,18–20], have

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attracted special interest during the last few years. For the aprotic IL 1-ethyl-3-methyl-imidazolium acetate confined between glass plates, a rapid and spontaneous reaction takes place leading to the formation of a neutral chemical moiety, which was assigned as N-heterocyclic carbene [17]. Enhanced diffusivity of cations has been observed for alkylammonium [15,21] and 1-ethyl-3-methyl-imidazolium [20] nitrates confined in layers between glass plates. Magnetic field strength is an important external variable that may influence the physical phase state and dynamic properties of some liquid systems, such as liquid crystals with molecules having anisotropic magnetic susceptibility [22], and solid materials, such as organic-based semiconductors [23]. There are reports on magnetic ionic liquids (MILs), which contain ferromagnetic ions in their chemical structure. However, no perceptible magnetic field effects have been observed for non-magnetic ionic liquids [24,25] until it was found recently that self-diffusion and NMR relaxation of protic alkylammonium nitrates and aprotic 1-ethyl-3-methyl-imidazolium nitrate confined between glass plates reversibly alter after sample placement in a strong static magnetic field [18,20,21,26]. The main factors responsible for this effect are hydrogen-bond networking and the polarity of the surface. It has been suggested that the processes influencing the dynamics of EAN in this confinement are the phase transformations of the ILs [18,21]. It has also been demonstrated that the process can be described well by the Avrami equation, which is typical for autocatalytic processes [20,21]. The transition is stopped by freezing the sample, while cooling and heating investigations showed differences in freezing and melting behavior of the sample that had been either exposed or not exposed to the static magnetic field [21]. Mechanisms of enhanced diffusivity of ILs in confinement have been analyzed earlier for nanopores [13], where the phenomenon was explained by the loose packing of the ions. Evidently, this mechanism does not work for ILs placed between glass plates, where the mean inter-planar distance is on the order of a few micrometers [15]. Earlier, we suggested that the responsible mechanism occurs through changes in the H-bonding network due to the presence of charged silicon oxide surfaces [15,18,20]. Hydrogen bonding is known to lead to certain shifts in the NMR spectra [7,9]. Therefore, in this work we applied ^{1}H NMR spectroscopy to study temperature changes of chemical shifts of protons of confined EAN. The temperature dependence of diffusivity of the ethylammonium (EA) cation of EAN was studied as well.

2. Materials and methods

2.1. Sample preparation

Fig. 1 shows the chemical structures of ions of ethylammonium nitrate. EAN was synthesized and characterized at Chemistry of Interfaces of Luleå University of Technology [15]. The content of water in the synthesized EAN was less than 0.05 wt% as determined by Karl-Fisher titration (Metrohm 917 Karl Fisher Coloumeter with HYDRANAL reagent).

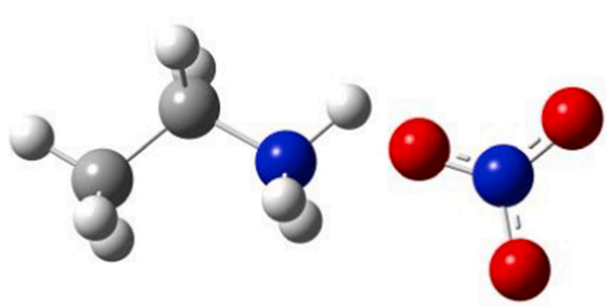

Fig. 1. Chemical structure of EAN: the ethylammonium cation (left) and the nitrate anion (right) with nitrogen (blue), oxygen (red), carbon (dark grey) and hydrogen (light grey) atoms shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

NMR measurements for the bulk EAN were performed by placing 300 μL of the IL in a standard 5-mm NMR glass tube. Fused quartz glass plates (UEG [Optics] Ltd. The Norman Industrial Estate, Cambridge, GB) with sizes 14 × 2.5 x 0.1 mm were used. The plates were thoroughly cleaned before sample preparation [15]). Confined ILs were prepared with glass plates settled in a stack (see Fig. S1 in the Electronic Supporting Information, ESI). A stack of glass plates filled with the IL was prepared in a glove box in a dry N_2 atmosphere. Samples were prepared by dropping 2 μL of the IL onto the first glass plate, placing another glass plate on top, adding 2 μL of the IL on top of this glass plate, etc. until the thickness of the stack reached 2.5 mm. Excess of the IL that overflowed at the edges of the stack was removed by sponging. The sample consisting of a stack of glass plates was positioned in a rectangular glass tube and sealed. The mean distance (spacing) between glass plates was estimated by weighing the introduced EAN and by measuring size of the stack of the glass plates before and after introducing the EAN. This gave a $d \sim 3.8$–4.5 μm. A detailed report of the sample preparation and characterization has been described in ESI of our previous papers [15,18]. The average glass plates spacing value agreed with the diffusion-time behavior of the diffusion decay form as obtained at measurements normal to the glasses [15]. To allow equilibration of ILs inside the samples, experiments were performed in a time interval from a week but not later than one and a half months after the sample preparation. It was stated earlier that EAN confined between glass plates remained either in an "**I**" phase in the absence of the magnetic field or in "$\mathbf{I_{MF}}$" phase formed after a long (~24 h) exposure to the magnetic field [21]. These phases differ in diffusion coefficients of EA: $\sim 7 \cdot 10^{-11}$ m^2/s for **I** and $\sim 3 \cdot 10^{-11}$ m^2/s for $\mathbf{I_{MF}}$, respectively, at 293 K. Therefore, phase state of confined EAN prior to the experiments was ensured by its diffusion coefficient at 293 K.

2.2. ^{1}H NMR spectroscopy

NMR experiments were performed on a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer. The working frequency for ^{1}H NMR was 400.21 MHz. The NMR spectra were recorded using a simple single $\pi/2$ radiofrequency pulse (pulse duration 7 μs). All resonance lines in the ^{1}H NMR spectra were referenced using tetramethylsilane (TMS).

2.3. Pulsed-field gradient NMR diffusometry

^{1}H NMR self-diffusion measurements were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a resonance frequency of 400.27 MHz for ^{1}H (magnetic field strength of 9.4 T with magnetic field homogeneity better than $4.9 \cdot 10^{-7}$ T) using a Diff50 Pulsed-Field-Gradient (PFG) probe. The homogeneity of the magnetic field was estimated from the ^{1}H NMR line width. An NMR solenoid ^{1}H insert was used to macroscopically align the plates of the sample stack at 0 and 90 degrees with respect to the direction of the external magnetic field (the same direction as that of the PFG). In the measurements, 90 degrees orientation of the normal to the glass plates in respect to the static magnetic field was used.

The diffusional decays (DD) were recorded using the spin-echo (SE, at $t_d \le 5$ ms) or the stimulated echo (StE) pulse trains. For single-component diffusion, the form of the DD can be described as [27,28,29]:

$$A(\tau, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2}\right)\exp(-\gamma^2\delta^2 g^2 D t_d) \tag{1a}$$

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right)\exp(-\gamma^2\delta^2 g^2 D t_d) \tag{1b}$$

for the SE and StE, respectively. Here, A is the integral intensity of the NMR signal, τ is the time interval between first and second radiofrequency pulses (SE and StE), τ_1 is the time interval between second and third radiofrequency pulses (StE). γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta / 3)$ is the diffusion time; Δ is the time interval between two identical gradient pulses. D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was 7 μs, δ was in the range of (0.5 ÷ 2) ms, τ was in the range of (3 ÷ 5) ms, and g was varied from 0.06 up to the maximum of the gradient amplitude, 29.73 $T{\cdot}m^{-1}$. Diffusion time t_d was varied from 4 to 500 ms. The repetition time during accumulation of signal transients was 3.5 s. The angle between the plates and the gradient direction was mounted by eye.

As has been shown in our previous work [15], restrictions had no effect on the D of EAN in the direction along the plates, while for diffusion normal to the plates, the effect of the plates on D was noticeable at t_d longer than 3 ms. For diffusion along the plates, D values were obtained by fitting Eq. (1) to the experimental decays. Data were processed using Bruker Topspin 3.5 software. Non-linear least squares regression was used to fit the experimental data with Eqs. (1) to extract D values.

It is known that NMR diffusion measurements at elevated temperatures can be distorted due to the temperature gradient inside the sample leading to convection [30]. The magnitude of this effect especially is pronounced in low-viscous liquids. It increases with the increase of temperature and length of the sample. To avoid convection in the bulk EAN, we used a sample with a length ~ 1 cm and Teflon insert the same as in our earlier paper [15]. The length of samples with confined EAN was 1.5 cm and the confinement generally hinders the convection. The presence of convection leads to the appearance of oscillation in diffusion decays and to increase of apparent diffusion coefficient with increasing diffusion time [27]. We used these features to control the convection effect in our measurements. No such effects were observed in the whole temperature range of measurements.

3. Results and discussion

3.1. Temperature dependences of ^{1}H NMR chemical shifts

Fig. 2 shows a ^{1}H NMR spectrum with assignments of lines of bulk EAN at 293 K. The resonance lines are broader in comparison with those of EAN dissolved in chloroform (Fig. S2 in the ESI). ^{1}H chemical shifts of NH_3 ($\delta[NH_3]$), CH_2 ($\delta[CH_2]$) and CH_3 ($\delta[CH_3]$) of EAN in the bulk phase at 293 K agree with literature data [31] and with our earlier obtained data [15]. Increasing temperature leads to minor changes in the chemical shifts of each group (Fig. S3 in the ESI). Tang et al. [9] observed much higher chemical shift variation of N—H protons of *n*-butylammonium acetate and *n*-butylammonium nitrate ILs in comparison with protons of alkyl groups. In our case of EAN, there is not such a great difference, therefore, the chemical environment of the protons, as well as the hydrogen bonding do not change much in comparison with n-butylammonium acetate at increasing temperature.

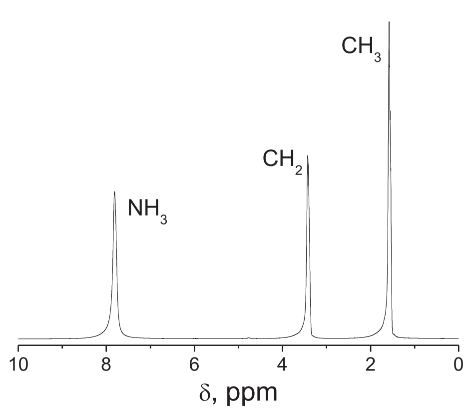

Fig. 2. ^{1}H NMR spectrum of bulk EAN. T = 293 K.

Placing EAN between glass plates did not significantly change the line widths of CH_2 and CH_3 groups, but increased the line width corresponding to exchangeable NH_3 protons [15]. Chemical shifts of protons for this system at 293 K are almost the same as in the bulk [15]. It has been shown that the dynamics of EAN placed between glass plates gradually changes after placement of the sample in a magnetic field [18]; however, the change is rather slow and takes a few hours to complete at room temperature. As the phase transition [21] is completed, the confined EAN remains in equilibrium and the NMR experiment can be performed in the same way as with the bulk EAN sample. In further discussion, we will use the same notations for dynamic EAN phases as we used earlier [21]: "$\mathbf{I_B}$" is the bulk phase, "**I**" is the phase formed in confinement in the absence of the magnetic field and "$\mathbf{I_{MF}}$" is the phase formed after a long (~24 h) exposure to the magnetic field.

We performed an experiment with the $\mathbf{I_{MF}}$ phase. To do this, after placement of the sample with confined EAN in the magnetic field of the NMR spectrometer, the sample was exposed there for 24 h, which was long enough to complete the phase transition $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ [18,21], and then the ^{1}H NMR spectrum was measured. Then the sample was heated stepwise by 10 K increments, equilibrated for 10 min and the ^{1}H spectrum was measured again, and so on, until the highest temperature of the experiment, 363 K, was reached. Dependences of $\delta(NH_3)$ and $\delta(CH_2)$ of confined EAN on temperature are shown in Fig. 3, with open and solid triangles, respectively, and in Table S1 in ESI. The dependence of $\delta(CH_3)$ on temperature showed the same peculiarity as $\delta(CH_2)$ and is not shown in the figure. We observed that the values $\delta(NH_3)$ and $\delta(CH_2)$ of confined EAN in the $\mathbf{I_{MF}}$ phase are slightly (~0.04 ppm) higher than those of bulk EAN (dotted and solid lines, respectively), while the trend in their change is the same as in the $\mathbf{I_B}$ phase.

The type of experiment performed for the sample in the **I** phase, which was not initially exposed to the strong static magnetic field of the NMR spectrometer, occurs under non-equilibrium conditions because of the continuously changing dynamics [18] and the phase's contents [21] of the confined EAN. Just after placement of the sample in the magnetic field at 293 K (the room temperature), the ^{1}H NMR spectrum was measured that takes around 3 min. Then the sample was heated stepwise by 10 K, equilibrated for ten minutes and the ^{1}H spectrum was measured until a temperature of 363 K was reached. Dependences of $\delta(NH_3)$ and $\delta(CH_2)$ of confined EAN that initially persisted in phase **I** are shown in Fig. 3 with open and solid circles, respectively. Again, the dependence of $\delta(CH_3)$ showed the same peculiarity as that of $\delta(CH_2)$ and is not shown in the figure. Values for $\delta(CH_3)$ and $\delta(CH_2)$ (solid circles) obtained in this experiment are in between the corresponding chemical shifts of bulk EAN (solid line) and confined EAN in the $\mathbf{I_{MF}}$ phase (solid triangles), while their temperature trends are close to those obtained for $\mathbf{I_B}$ and for $\mathbf{I_{MF}}$. In all cases, slight (~0.0015 ppm/K) and similar downfield shifts of $\delta(CH_3)$ and $\delta(CH_2)$ occur with increasing temperature. At the same time, heating the $\mathbf{I_B}$ and $\mathbf{I_{MF}}$ phases leads to a slight (~0.00076 ppm/K), gradual upfield shift of $\delta(NH_3)$ (dotted line and open triangles). Similar shifts have been observed earlier [9,32], where this effect has been related to H-bond weakening and, therefore, lessening of the electron-withdrawing effect of the hydrogen bond acceptor on the proton with an increase in temperature. As a result, the proton becomes more shielded and its chemical shift value decreases. Indeed, the chemical shift is a measure of electron density near the proton being measured.

Therefore, we suggest that the observed trend of chemical shifts of $\delta(NH_3)$ in these experiments is due to H-bond weakening and at the same time increased contributions of other types of interactions, such as ionic and wan-der-Waals interactions, occurring in the $\mathbf{I_B}$ and $\mathbf{I_{MF}}$ phases with rising temperature. Alkyl protons do not take part in forming hydrogen bonds with anions directly.

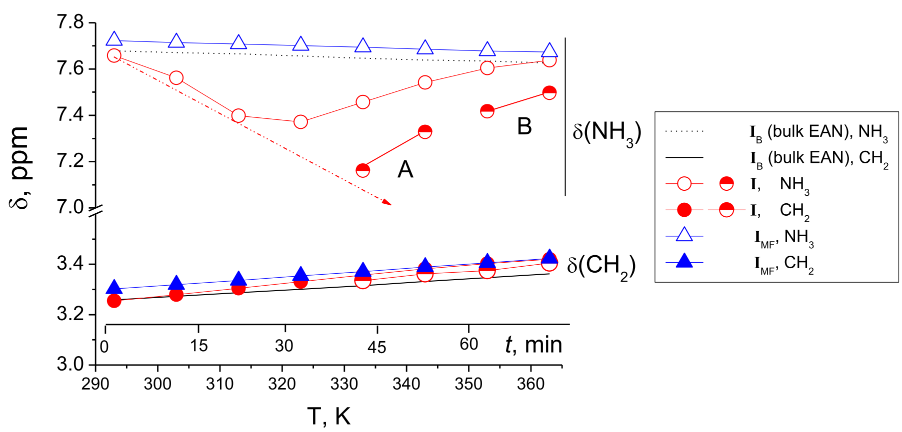

Fig. 3. Temperature dependences of [1]H NMR chemical shifts of different chemical groups of EAN in bulk, EAN confined between glass plates in the $\mathbf{I_{MF}}$ phase (triangles) and EAN experiencing $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ transition (circles). Dependences for confined EAN were obtained in the process of stepwise increases in temperature by 10 K, equilibration for ten minutes and NMR spectra measurement for one minute. Axis "t" shows the timing of experiments with the confined EAN. Confined EAN samples A and B were placed in the magnet field of the NMR spectrometer at temperatures of 333 K and 353 K, respectively. Further explanation is given in the text.

However, the most intriguing is the change of δ(NH_3) for EAN in the sample, which was initially in the **I** phase and experienced dynamic and phase transformation transition $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ in the magnetic field simultaneously with stepwise heating (open circles). Raising the temperature leads to a significant upfield shift of δ(NH_3) in the temperature range of 293–313 K, with a further downshift of δ(NH_3) at further rise of temperature. From the time dependence of the EA cation diffusion coefficient at 293 K [18], it is known that major changes in cation dynamics occur at times longer than 20 min and the whole $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ transformation continued during around four hours at 293 K. Therefore, the first three points of this dependence in the low-temperature range correspond to the temperature dependence of δ(NH_3) in the almost undisturbed **I** phase. Further exposure to the magnetic field and the possible effect of higher temperature leads to an increase in the contribution of the $\mathbf{I_{MF}}$ phase with corresponding values of δ(NH_3). To check this suggestion, we performed two additional experiments. Two samples with confined EAN in the **I** phase (not exposed previously to the magnetic field) were placed in the NMR spectrometer probe at 333 K (sample A) and 353 K (sample B), equilibrated with the probe temperature for ten minutes and the [1]H NMR spectra were measured. Then the temperature was raised by ten degrees, the sample was equilibrated again for ten minutes and the [1]H NMR spectra were measured again. In the experiments, we tried to obtain less disturbed chemical shifts corresponding to the **I** phase at higher temperatures in the studied temperature range. δ(CH_3) and δ(CH_2) values (half-filled circles), as expected, were close to those obtained for the $\mathbf{I_{MF}}$ phase (solid triangles). Values of δ(NH_3) are shown in Fig. 3 as half-filled circles. Indeed, comparing data for samples A and B (half-filled circles) with δ(NH_3) obtained for the sample initially placed in the magnetic field at 293 K and studied in the whole range of time and temperatures of the experiment (open circles), we can see effects of both, time and temperature on the δ(NH_3) values. The shorter time of exposure to the magnetic field for sample A, which was initially in the **I** phase (ten minutes at T = 333 K) leads to a stronger upfield shift of δ(NH_3). For sample B, which was also exposed to the magnetic field for ten minutes, but at 353 K, the shift is less pronounced. We suggest that the increase in temperature accelerates the transformation $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ of the confined EAN. Based on the results obtained, we can propose a tentative dependence of δ(NH_3) for the confined EAN in the **I** phase, as shown by a dashed-dotted line in Fig. 3. Generally, the effect of temperature on δ(NH_3) of confined EAN in the **I** phase is much higher than in bulk and in confined EAN in the $\mathbf{I_{MF}}$ phase. This dependence corresponds to nearly 0.0145 ppm/K, which is at least double that observed previously for other ILs [9].

3.2. Temperature dependences of diffusivities

Temperature dependences of diffusion coefficients (*Ds*) of the EA cation of EAN in bulk ($\mathbf{I_B}$ phase) and in $\mathbf{I_{MF}}$ phase were obtained through the process of increasing the temperature, in the same way as for chemical shift experiments described above. For bulk EAN, the temperature range was limited in the low temperature range due to EAN crystallization at 285 K. For confined EAN in the $\mathbf{I_{MF}}$ phase, the temperature range was 253–333 K, while for confined EAN in the **I** phase, the temperature dependence was obtained only in the range of 293–313 K, according to the chemical shift experiment. To prevent transformation $\boldsymbol{I} \rightarrow \boldsymbol{I_{MF}}$, newly prepared samples that remained in the **I** phase were used. The sample was placed in the NMR probe at a certain temperature, equilibrated there for ten minutes and the NMR *D* was measured for nearly one minute as we have done earlier [18]. Variations of SE and StE sequences were applied for the measurements with different t_d values. Deviations of forms of diffusion decay from exponentiality (Eqs. 1a,b) were examined that might be a sign of possible effects of molecular exchange between phases as well as repulsion of molecules from phases boundaries. The measurement showed that the obtained diffusion decays were single-componential and diffusion coefficients do not depend on t_d. Temperature dependences of representative values of diffusion coefficients in bulk and confined EAN are shown in Fig. 4 in Arrhenius coordinates and in Table S2 in the ESI. These temperature dependences are not strictly linear. An Arrhenius function shows a linear dependence in Arrhenius coordinates. This is not the case for our system, as seen in Fig. 4. Non-Arrhenius dependences of diffusion coefficients are typical for ILs [33]. A more universal form of temperature dependences taking into consideration the proximity of the system temperature to their glass transition

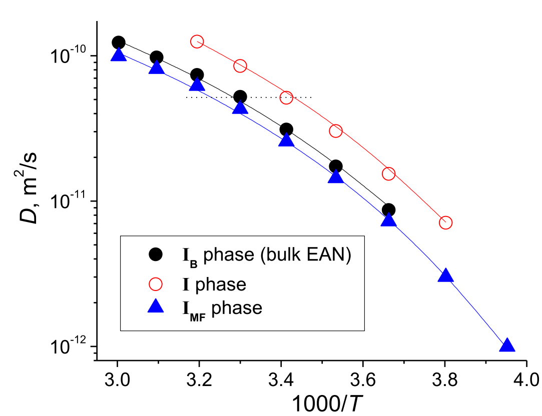

Fig. 4. Temperature dependences of diffusion coefficient of the ethylammonium cation of EAN in bulk ($\mathbf{I_B}$ phase, solid circles), and of EAN confined between glass plates in the **I** phase (open circles) and $\mathbf{I_{MF}}$ phase (triangles). Lines show best fits with the VFT equation. Induction of the magnetic field was 9.4 T. SE pulse train: t_d = 4 ms, δ = 1 ms, g_{max} = 29.73 T·m^{-1}, acquisition time 0.15 s, spectral width 39.7 kHz, number of scans 16.

Table 1
VFT equation fitting parameters and apparent energies of activation for diffusion data for bulk EAN ($\mathbf{I_B}$) and confined EAN in **I** and $\mathbf{I_{MF}}$ phases (Fig. 4).

Phase	D_0*10^{-9} m^2/s	B	T_0, K	E_D, kJ/(mol·K)
$\mathbf{I_B}$	4.0	480	194	4.0
I	6.5	470	194	3.9
$\mathbf{I_{MF}}$	3.3	480	194	4.0

temperature, T_0, is the Vogel-Fulcher-Tammann (VFT) equation in the form for diffusivity [33]:

$$D = D_0 \exp\left(\frac{-B}{(T - T_0)}\right) \tag{2}$$

where D_0, T_0 and B are adjustable parameters. Apparent energy of activation for diffusion is related to B as $E_D = B{\cdot}R$. We have described D (T) in Fig. 4 by fitting D_0, T_0 and B. The procedure of fitting was performed in two steps. In the first step, we plotted ln(D) vs 1/(T-T_0) and selected T_0 to have this dependence be linear. Uncertainty in this step was ± 2 K. In the second step, we fit the dependence by a linear regression to obtain fitting parameters (D_0, B). Solid lines in Fig. 4 show the best results of the fitting and the corresponding fitting parameters are presented in Table 1.

VFT analysis is rather formal, but it may show which parameters are responsible for changes occurring with a substance at a certain transformation. In our case, the mechanism of thermal activation is the same for all three phases, $\mathbf{I_B}$, **I** and $\mathbf{I_{MF}}$, as is evident from the fact that all three share the same value of the energy of activation E_D. T_0 also does not change significantly. The main change in diffusivity is due to the change in D_0. This is in agreement with changing the density of H-bonding, which occurs when placing EAN in confinement between polar glass (quartz) plates ($\mathbf{I_B} \rightarrow \mathbf{I}$ transition) and exposing it to a magnetic field ($\mathbf{I} \rightarrow \mathbf{I_{MF}}$ transition).

Diffusion coefficients in bulk EAN ($\mathbf{I_B}$ phase) and confined EAN in the **I** and $\mathbf{I_{MF}}$ phases increase with a rise in temperature in a concerted way (Fig. 4). Values of Ds follow the order $D(\mathbf{I}) > D(\mathbf{I_B}) > D(\mathbf{I_{MF}})$ in the whole temperature range. This order characterizes the thermal energy of the EA cation in different phases and its difference is related, evidently, to the difference in the H-bonding network in these phases: the H-bonding network is weaker in the **I** phase and stronger in the $\mathbf{I_{MF}}$ phase. Therefore, confinement of EAN between the plates ($\mathbf{I_B} \rightarrow \mathbf{I}$) is an endothermic process, while processes occurring in the magnetic field, $\mathbf{I} \rightarrow \mathbf{I_{MF}}$ and $\mathbf{I_{MF}} \rightarrow \mathbf{I}$, are exothermic and endothermic, respectively. A loss of energy of thermal motion at any temperature in the $\mathbf{I_B}$ and $\mathbf{I_{MF}}$ phases in comparison with the **I** phase can be compensated for by heating the sample. For example, the same diffusion coefficient $5.13{\cdot}10^{-11}$ m^2/s obtained at 293 K in the **I** phase can be obtained by heating the $\mathbf{I_B}$ phase to 303 K or the $\mathbf{I_{MF}}$ phase to 309 K (dotted line in Fig. 4). The increase of thermal energy ΔE of the **I** phase relative to $\mathbf{I_B}$ phase is:

$$\Delta E = \frac{3}{2}R(T_{I_B} - T_I) \tag{3}$$

Such calculations give $\Delta E \sim 120$ J/mol and ~ 230 J/mol for transitions $\mathbf{I} \rightarrow \mathbf{I_B}$ and $\mathbf{I} \rightarrow \mathbf{I_{MF}}$, respectively. These differences in energy characterize translational energy gains due to the decreased H-bonding density in the **I** phase. Therefore, placing and removing the sample with EAN confined between glass plates in a static magnetic field allows taking and release energy in the energy of H-bonding. Until now, no satisfactory theory or model have been able to explain the effect of H-bonding on ILs. It is known that in imidazolium ILs hydrogen bonding leads to higher-viscosity liquids and higher melting points [32]. This generally agrees with our data: higher diffusivity is observed in the **I** phase with less dense H-bonding where we can expect less viscosity η, because of the relation $\eta \sim 1/D$. The melting point in the **I** phase is lower than that in the $\mathbf{I_B}$ and $\mathbf{I_{MF}}$ phases [21].

Therefore, the H-bonding network determines local chemical shielding, melting behavior and diffusivity in EAN. Confinement of EAN between glass plates weakens the network. It is known that water protons interact with the surface of glass in an exothermic process and form silanol bonds (Si-O-H) [34,35], while long-range SiO_2-water interaction is not important. Protons of the NH_3 group can also, similar to water protons, interact with the polar glass surface. In the case of EAN near the surface of mica, ordered layers may form near the solid support up to a distance of 10 nm [12], mainly due to ionic interactions. Evidently, modification of the H-bonding network in the layer of EAN confined between glass plates, without and under the influence of a static magnetic field, belongs to a different class of phenomena that currently are not fully understood.

4. Conclusion

The dynamic properties of ethylammonium nitrate protic IL are determined primarily by the hydrogen-bonding network between cations and anions, with a major contribution from NH_3 protons. Confinement of EAN between parallel glass (quartz) plates in a layer with a thickness of nearly 4 μm weakens this H-bonding network in the whole EAN layer, which leads to an upfield shift of the NMR chemical shift of NH_3 protons, a decrease in melting temperature of EAN, and an increase in the diffusivity of the IL. Exposure of the confined EAN to a static magnetic field leads to restitution of the H-bonding network, which can be observed by ^{1}H NMR chemical shift of NH_3 protons as well as diffusivity of the ethylammonium cation.

The mechanism of alteration of the H-bonding network in a confined layer of EAN is still unclear and remains under study in our laboratory.

Author statement

Oleg I. Gnezdilov: Methodology, Investigation, Data curation.
Oleg N. Antzutkin: Reviewing and Editing.
Rustam Gimatdinov: Software, Validation.
Andrei Filippov: Conceptualization, Investigation, Writing-Original draft preparation.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.mri.2020.09.012.

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